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Docket No.: 50-366

NL-23-0915

**U. S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D. C. 20555-0001**

**Edwin I. Hatch Nuclear Plant – Unit 2
Licensee Event Report 2023-003-00
Unit 2 Manual Reactor Trip due to Loss of Feedwater**

Ladies and Gentlemen:

In accordance with the requirements of 10 CFR 50.73(a)(2)(iv)(A), Southern Nuclear Operating Company hereby submits the enclosed Licensee Event Report.

This letter contains no NRC commitments. If you have any questions, please contact the Plant Hatch Licensing Manager, Jimmy Collins, at 912-453-2342.

Respectfully submitted,

A handwritten signature in black ink, appearing to read 'M. Busch'.

**M.S. Busch
Vice President – Hatch**

MSB/CJC

Enclosure: LER 2023-003-00

**Cc: NRC Regional Administrator, Region II
NRR Project Manager – Plant Hatch
NRC Senior Resident Inspector – Plant Hatch
RTYPE: CHA02.004**

**Edwin I. Hatch Nuclear Plant - Unit 2
Licensee Event Report 2023-003-00
Unit 2 Manual Reactor Trip due to Loss of Feedwater**

Enclosure

LER 2023-003-00

NRC FORM 366 (10-01-2023)		U.S. NUCLEAR REGULATORY COMMISSION			APPROVED BY OMB: NO. 3150-0104		EXPIRES: 03/31/2024			
LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/) Estimated burden per response to comply with this mandatory collection request: 60 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to InfoCollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; email: omb_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.										
1. Facility Name Edwin I. Hatch, Unit 2					☒ 050 ☐ 052		2. Docket Number 00366		3. Page 1 OF 2	
4. Title Manual Reactor Trip Due To Loss of Feedwater										
5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050 Docket Number
11	01	2023	2023	- 003 -	00	12	19	2023	Facility Name	☐ 052 Docket Number
9. Operating Mode 1						10. Power Level 056				
11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)										
10 CFR Part 20		☐ 20.2203(a)(2)(vi)	10 CFR Part 50		☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)			
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)					
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)					
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☒ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)					
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)					
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)					
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)					
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)					
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)							
☐ OTHER (Specify here, in abstract, or NRC 366A).										
12. Licensee Contact for this LER										
Licensee Contact Jimmv Collins - Licensing Manager								Phone Number (Include area code) 9124532342		
13. Complete One Line for each Component Failure Described in this Report										
Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS	
X	SJ	P	B580	Y						
14. Supplemental Report Expected					15. Expected Submission Date			Month	Day	Year
☒ No					☐ Yes (If yes, complete 15. Expected Submission Date)					
16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)										
On November 1, 2023, reactor power was being lowered on Unit 2 to support removal of the 'A' Reactor Feed Pump (RFP) from service due to a suction pipe drain line leak. The 'A' RFP was removed from service and isolated at 0642 EDT. During this evolution, water from the leak on the drain line of the 'A' RFP entered various electrical cabinets and components related to the 'B' RFP, causing the 'B' RFP to trip at 0648 EDT. The trip of the 'B' RFP resulted in a loss of feedwater and lowering of reactor water level. Due to the lowering reactor water level, operators manually inserted a reactor trip in anticipation of water level dropping to the automatic reactor protection system (RPS) trip setpoint. The Reactor Core Isolation Cooling (RCIC) and High Pressure Coolant Injection (HPCI) pumps were used to restore reactor water level. All safety system responded as required. The cause of the leak from the 'A' RFP suction drain line was a pipe shear due to excessive stresses from thermal cycling and flow induced vibration. This was corrected by implementing a design change that cut and capped that portion of piping. The cause of the 'B' RFP trip was electrical grounding due to water intrusion from the 'A' RFP leak resulting in an invalid low suction pressure trip on the 'B' RFP.										

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; email: oir_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME		☒ 050 ☐ 052	2. DOCKET NUMBER	3. LER NUMBER		
Edwin I. Hatch, Unit 2			00366	YEAR	SEQUENTIAL NUMBER	REV NO.
				2023	003	00

NARRATIVE**EVENT DESCRIPTION**

On November 1, 2023, reactor power was being lowered on Unit 2 to support removal of the 'A' Reactor Feed Pump (RFP) [EIS Code: SJ] from service due to a leak on a drain line connected to the suction piping of the pump. At 0642 EDT with the reactor at approximately 56% rated thermal power, the 'A' RFP was removed from service and isolated. While removing the 'A' RFP from service, water from the leak on the drain line of the 'A' RFP entered various electrical cabinets and components related to the 'B' RFP, causing the 'B' RFP to trip at 0648 EDT. The trip of the 'B' RFP resulted in a loss of feedwater and reactor water level lowering. Due to the lowering reactor water level, operators manually inserted a reactor trip in anticipation of water level dropping to the automatic trip setpoint. The Reactor Core Isolation Cooling (RCIC) [EIS Code: BN] and High Pressure Coolant Injection (HPCI) [EIS Code: BJ] pumps automatically started and were used to restore reactor water level. Secondary containment isolated and Standby Gas Treatment (SBGT) [EIS Code: BH] autostarted, and Group 2 and Group 5 primary containment isolation valves [EIS Code: JM] isolated as expected when their setpoints were reached. There was no safety related equipment inoperable at this time and no other nonfunctional, non-safety equipment that contributed to this event.

This event is reportable under 10 CFR 50.73(a)(2)(iv)(A) as an event that resulted in manual actuation of the reactor protection system and automatic actuation of general containment isolation signals affecting containment isolation valves in more than one system. Additionally, HPCI and RCIC automatically started when their reactor water level setpoints were reached.

EVENT CAUSE ANALYSIS

A root cause analysis was performed and determined that the direct cause of the leak from the 'A' RFP suction drain line was a pipe shear due to excessive stresses from thermal cycling and flow induced vibration. The cause of the 'B' RFP trip was electrical grounding due to water intrusion from the 'A' RFP leak resulting in an invalid low suction pressure trip on the 'B' RFP.

SAFETY ASSESSMENT

There were no safety consequences as a result of this event. All safety related systems functioned as designed during this event. The operating crew responded correctly to the event. The applicable abnormal/emergency operating procedures were entered. The event was within the analysis of the Updated Final Safety Analysis Report (UFSAR) Chapter 15. No radiological release occurred due to this event.

CORRECTIVE ACTION

The leak from the 'A' RFP suction piping drain line was corrected by implementing a design change that cut and capped that portion of piping. The cause of the 'B' RFP trip was electrical grounding due to water intrusion from the 'A' RFP leak resulting in an invalid low suction pressure trip on the 'B' RFP. All electrical components related to the 'B' RFP were investigated. Impacted components were dried out or replaced as required.

PREVIOUS SIMILAR EVENTS

None.